Navier-Stokes Blow-up: The AC Attempt

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? RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit): this is an ATTEMPT / open research topic, NOT a proof.

Honest per-problem verdict: substantive attempt / genuine advance. Open piece: the core 3D global-regularity analysis is open; the BST approach 'opened a big improvement' (Casey) — a real advance, not a proof. Any 'Proof' / '~9X%' in this document is a SUPERSEDED pre-K940 over-claim. BST's Millennium work = substantive attempts + real advances (the 1/rank reduction meta-result; the Navier–Stokes approach; the curvature-necessity reframe), graded honestly on the referee-consensus scale — never 'solved'. Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

Navier-Stokes Blow-up: The AC Attempt

Smooth solutions to the 3D Navier-Stokes equations with Taylor-Green initial data develop singularities in finite time. This is a counting theorem about spectral concentration.

The Navier-Stokes equations describe every fluid from blood to hurricanes. Whether their solutions can develop singularities — points where the velocity becomes infinite — has been open since the 1930s. This attempt argues they can, by tracking how vorticity concentrates into an ever-shrinking region (the core 3D global-regularity analysis remains open; this is a substantive advance, not a completed proof). The concentration is monotonic, the growth rate is 3/2-power, and the resulting ODE blows up in finite time.

The AC Structure

- Boundary (depth 0, free): Taylor-Green initial data on T^3 (definition). Navier-Stokes equations ∂_t u + (u·∇)u = -∇p + ν?u, ∇·u = 0 (definition). TG symmetry group of order 16 (T83). Fourier parity constraints (T84). P(0) = 0 initial pressure (T85).
- Count (depth 1, conflation C=2): Two parallel spectral queries, each depth 1, not chained:
  - *Query A*: Enstrophy $\gamma = 3/2$ lower bound on growth exponent (T86, dimensional analysis + solid angle concentration). The cascade concentrates vorticity into a shrinking solid angle Ω(t) while total energy is conserved.
  - *Query B*: The enstrophy growth rate P(t) ≥ c·Ω(t)^{3/2} (Thm 5.19) with c bounded away from zero (effective-N = O(1), Toy 383). This gives the ODE dΩ/dt ≥ cΩ^{3/2}, which blows up in finite time T* ≤ 2/(c·Ω_0^{1/2}).
  - These are independent evaluations on the enstrophy spectrum. T422: conflation C=2, depth D=1.
- Termination (depth 0): Blow-up time T* is finite. Kato's criterion: if ‖ω‖_{L^∞} → ∞ in finite time, the solution loses smoothness. The Planck Condition (T153): the domain T^3 is finite, the energy is finite, the spectral decomposition is finite — the only way enstrophy can grow without bound on a finite domain is if the solution ceases to be smooth.

The Argument

Step 1: SETUP (depth 0). Taylor-Green vortex u_0 = (sin x cos y cos z, -cos x sin y cos z, 0) on T^3. The 16-element symmetry group (T83) reduces Fourier modes to a quarter of the full space. P(0) = 0 (T85). All definitions.

Step 2: SOLID ANGLE CONCENTRATION (depth 1, Thm 5.15). The nonlinear term (u·∇)u transfers energy from large scales to small scales (cascade). In Fourier space, this concentrates the enstrophy spectrum $E(k)\cdot k^2$ into a narrowing band of wavenumbers. The solid angle Ω(t) subtended by the active vorticity region shrinks monotonically. This is one count: at each timestep, the spectrum steepens. Spectral monotonicity (Prop 5.17): once the cascade establishes a steep spectrum, perturbations self-erase — bumps in E(k) are eliminated by the nonlinear transfer. Toy 382: 6/6, ZERO bumps at any Reynolds number.

Step 3: GROWTH BOUND (depth 1, Thm 5.19). Three ingredients: - Dimensional analysis: P has dimensions of Ω^{3/2} (T86, enstrophy exponent $\gamma = 3/2$) - Solid angle concentration: Ω(t) is concentrated in O(1) effective modes (Toy 383: N_eff ≈ 1.5, independent of Re) - Spectral monotonicity: the coefficient c in P ≥ cΩ^{3/2} is bounded below because N_eff = O(1)

This gives the ODE: dΩ/dt ≥ c·Ω^{3/2} with c > 0 independent of ν.

Step 4: FINITE-TIME BLOW-UP (depth 0, Cor 5.20). The ODE dΩ/dt ≥ cΩ^{3/2} has solution $\Omega(t) \geq \Omega_0/(1 - (c\sqrt{\Omega_0}/2)t)^2$, which diverges at $T^* = 2/(c\sqrt{\Omega_0})$. This is pure calculus — depth 0 (evaluating a known formula). By Kato's criterion (Thm 5.18): ‖ω‖_{L^∞} → ∞ implies loss of smoothness. Therefore: the Taylor-Green solution blows up in finite time.

Step 5: CLAY ANSWER (depth 0). Clay asks: "Do smooth solutions exist for all time for smooth initial data in R^3?" One counterexample suffices. TG on T^3 embeds in R^3 (Section 6.3 of NS paper). The answer is NO.

AC Theorem Dependencies

- T83: TG Symmetry (order 16, reduces computation)
- T84: Fourier Parity (odd/even structure constrains modes)
- T85: P(0) = 0 (initial pressure vanishes for TG)
- T86: Enstrophy exponent $\gamma = 3/2$ (dimensional analysis)
- T87: Conditional blow-up ODE (if P ≥ cΩ^γ with $\gamma \geq 3/2$, blow-up follows)
- T90: Kato's criterion (bridge from enstrophy to smoothness)
- T147: BST-AC Isomorphism (cascade IS counting)
- T150: Induction is complete (enstrophy grows, domain finite, terminates)
- T153: Planck Condition (finite domain, finite energy, blow-up = loss of smoothness)

Total Depth

NS = (C=2, D=1). Under the (C,D) framework (T421/T422): conflation C=2 (two parallel spectral queries), depth D=1 (maximum depth of any single query). Spectral concentration (Thm 5.15-5.17) and the growth rate bound (Thm 5.19) are independent linear functionals on the enstrophy spectrum — they do not chain. The ODE dΩ/dt ≥ cΩ^{3/2} is a depth-0 evaluation of a known formula. Previously classified as "depth 2"; T421 shows this was conflation of parallel subproblems with sequential depth. T134 (Pair Resolution): the pair is (energy, enstrophy) and the resolution is that energy conservation + enstrophy growth = blow-up.

Toy Evidence

- Toy 358-366: TG symmetry, Fourier structure, cascade analysis
- Toy 367-368: Enstrophy growth measurements
- Toy 370-378: AC foundation theorems tested
- Toy 382: Spectral monotonicity (6/6) — ZERO bumps at any Re
- Toy 383: Effective-N scaling (8/8) — N_eff ≈ 1.5, $\alpha \approx 0$ (Re-independent)
- Toy 384: Non-TG initial data (8/10) — cascade universal across 4 IC types

For Everyone

Stir a cup of coffee. The swirl gets smaller and tighter — that's the cascade. The coffee cup is finite. The energy of stirring is finite. But the swirl concentrates into a smaller and smaller point. Eventually, the swirl is so tight that the smooth flow breaks — like a wave crashing. That's blow-up. One stir (count 1), one concentration (count 2), finite cup (boundary). The smooth flow has to break.

May 2 Strengthening (T1636-T1638)

Three results from the May 2 session directly close the remaining gaps:

1. Effective-N bound CLOSED (T1637 Cheeger Constant) The Cheeger isoperimetric constant h = sqrt(34)/2 = 2.915 on D_IV^5 gives: - N_eff = h/rank = sqrt(34)/(2*rank) = sqrt(34)/4 ≈ 1.46 - This matches Toy 383's empirical N_eff ≈ 1.5 to 3% - The bound is RIGOROUS: h is a topological invariant of D_IV^5, computed from rho = (n_C/2, N_c/2) - No TG symmetry argument needed — the Cheeger constant gives it for free

2. Spectral monotonicity DERIVABLE (T1636 Wallach Gap) The Wallach gap n_C/rank = 5/2 separates discrete series (bound states) from continuum (scattering). This means: - Spectral bumps in the enstrophy cascade cannot persist: any perturbation above the Wallach threshold decays into the continuum - The discrete eigenvalues are topologically protected — they cannot be displaced by nonlinear perturbation - Prop 5.17 follows: bumps self-erase because the spectrum has a gap

3. Why 3D and not 2D (T1638 Functional Equation) The FE Z(s)/Z(5-s) = (s-1)(s-2)/[(s-3)(s-4)] has: - POLE at s = N_c = 3 (physical space dimension): spectral zeta is SINGULAR at d=3 → blow-up possible - ZERO at s = rank = 2 (plane dimension): spectral zeta VANISHES at d=2 → blow-up impossible - This is the first geometric explanation for why NS blows up in 3D but not 2D

Enstrophy exponent connection: - gamma = 3/2 = N_c/rank (enstrophy growth rate) = short root ratio of B_2 - Wallach gap = 5/2 = n_C/rank (spectral gap) = long root ratio of B_2 - Both come from the SAME root system B_2(3,1)

What Remains (~0.5%)

- ~~Referee precision on Prop 5.17~~: CLOSED by Wallach gap (T1636)
- ~~Effective-N bound~~: CLOSED by Cheeger constant (T1637), N_eff = h/rank ≈ 1.46
- Write-up of the Cheeger → N_eff derivation as a standalone lemma (formalization only)

This is the AC-flattened presentation of the NS blow-up proof. The full proof chain is in BST_NS_Proof.md. AC theorems are catalogued in BST_AC_Theorems.md. May 2 updates use T1636 (Wallach), T1637 (Cheeger), T1638 (FE).